CSE 4128

LAB TEST: Section B

Dr. Sk. Md. Masudul Ahsan
Professor,
Dept. of Computer Science and Engineering,
KUET

Md Tajmilur Rahman
Lecturer,
Dept. of Computer Science and Engineering,
KUET

Problem 1

- Convolve the following Lena image by the given kernel by creating your own defined function for convolution
  - Encircled cell shows the center of the kernel

Input

$$\begin{bmatrix} -2 & -1 & 0 & 1 & 2 \\ -1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 2 & 1 & 0 & -1 \\ 2 & 1 & 0 & -1 & -2 \end{bmatrix}$$

Kernel

Sample output

Problem 2

Detect the edges of the given image using the Sobel operator and Double Thresholding.

1. Find the Gradient magnitude of the input image using the given kernels.
2. Apply double thresholding in the gradient magnitude to detect edges using the given equation.

$$g(x,y) = \begin{cases} 255 & if\ f(x,y) > Th \\ 0 & if\ f(x,y) \leq Tl \\ 128 & Otherwise \end{cases}$$

-1	0	+1
-2	0	+2
-1	0	+1

Gx

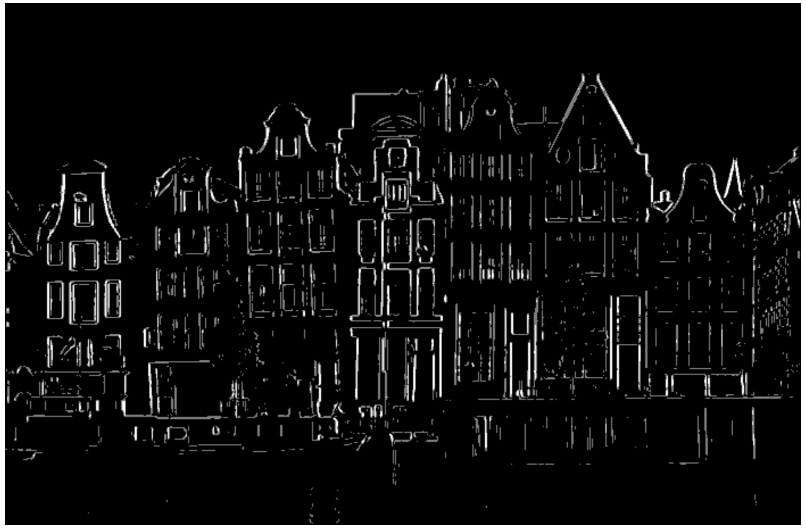
+1	+2	+1
0	0	0
-1	-2	-1

Gy

Sobel operator

N.B.: You can use any library functions

Input



Output (T_l = 100, T_h = 150)

Problem 3

Apply the following Notch pass filters on the Fourier domain to remove periodic noise from the image.

$$H(u,v) = \begin{cases} 1, & \text{if } D(u,v) <= D_0 \text{ and} \\ 0, & \text{if } D(u,v) > D_0 \end{cases}$$

Input

As for the center, use the given input values center **(X,Y)=(272, 256),and D_o=5**

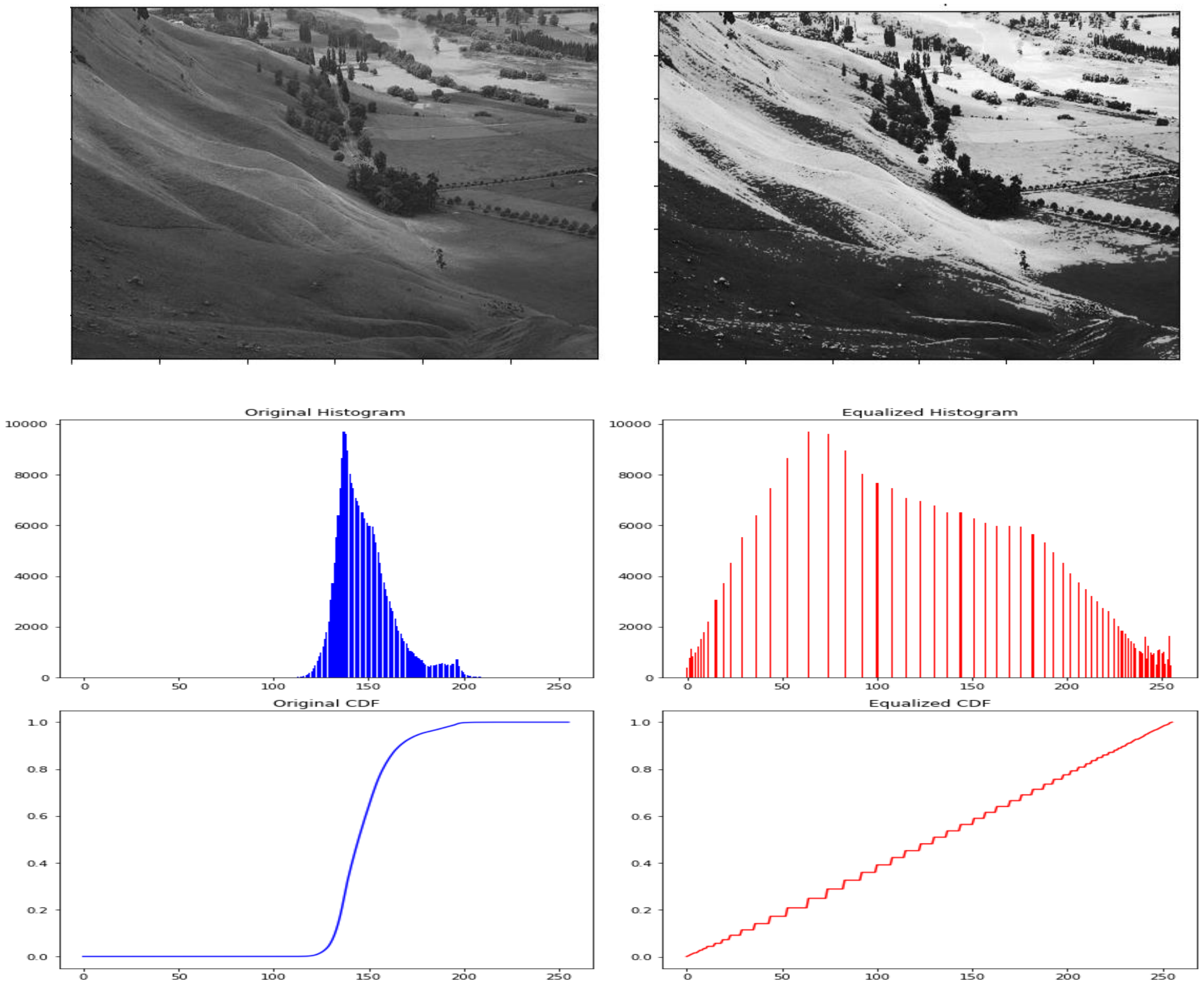
Problem 4

Stretch the image histogram so that its histogram is equalized over the whole intensity.

1. Show the output image after the equalization in the gray-scale image.
2. Generate the histogram and CDF of intensity of the input image and the equalized image.
3. Show the images, plot the pdf and cdf for both input and output images (side by side).

Input

Output



Problem 5

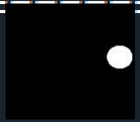
Calculate **Form factor, Roundness, and Compactness** for the given Binary images and calculate their **similarity** using given equation.

$$\textbf{Compactness} = \frac{(Perimeter)^2}{Area}$$

$$\textbf{Form factor} = \frac{4\pi * Area}{(Perimeter)^2}$$

$$\textbf{Roundness} = \frac{4 * Area}{\pi * MaxDiameter^2}$$

$$\boldsymbol{Distance(X,Y)} = \left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{\frac{1}{p}}$$

			
	33.01	106.78	630.53
	42.66	31.11	554.86
	568.94	495.18	28.68
	1643.85	1717.62	2241.38

Here, X and Y are the descriptor vectors for 2 images, n=3 for Form factor, Roundness, and Compactness. **Use p = 2 for the similarity measurement.**